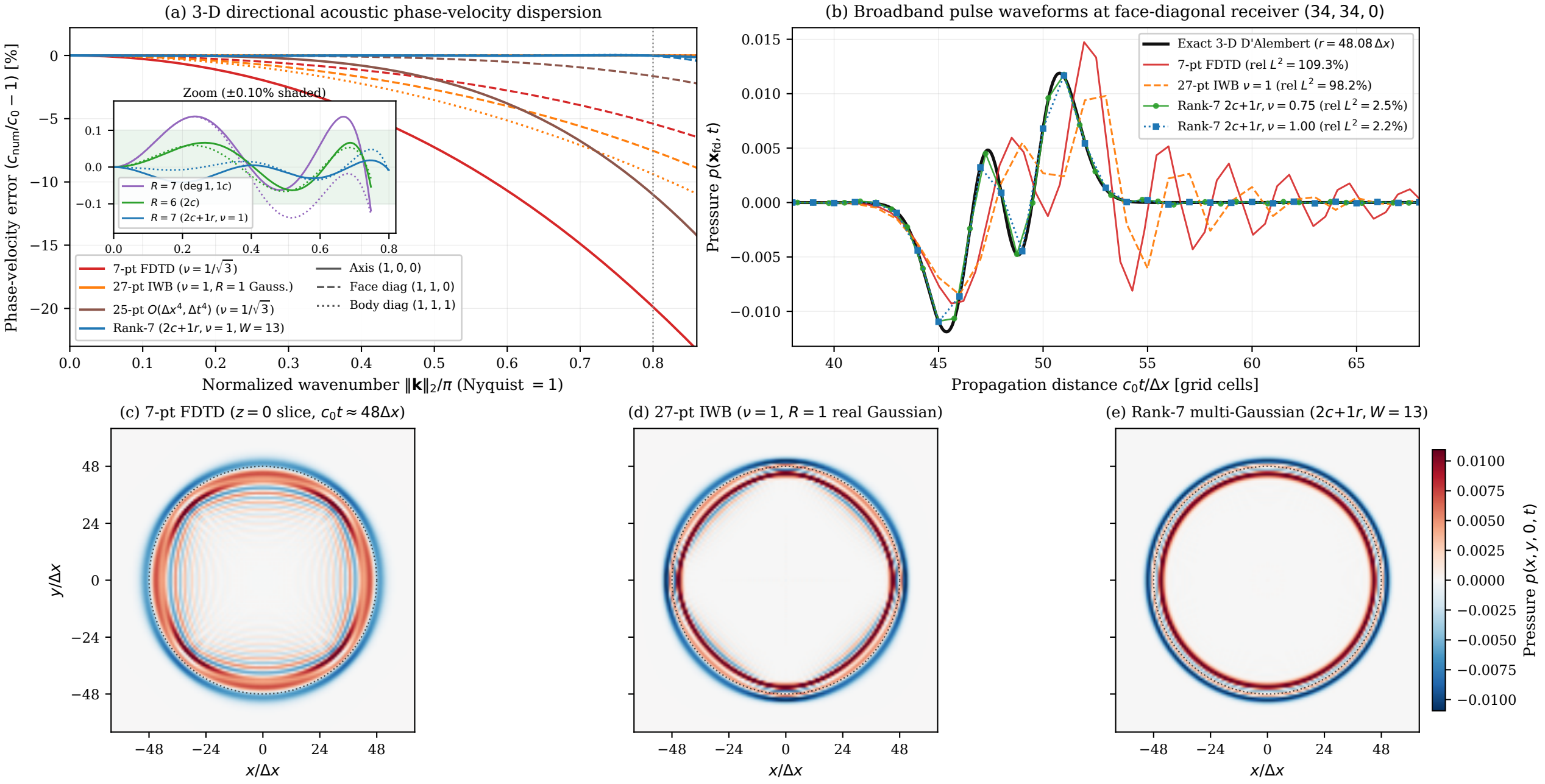

(a) 3-D directional acoustic phase-velocity dispersion
Phase-velocity error $(c_{num}/c_0 - 1)$ [%]
Normalized wavenumber $\|\mathbf{k}\|_2/\pi$ (Nyquist = 1)
Zoom (±0.10% shaded)
R = 7 (deg 1, 1c)
R = 6 (2c)
R = 7 (2c+1r, ν = 1)
7-pt FDTD (ν = 1/√3)
27-pt IWB (ν = 1, R = 1 Gauss.)
25-pt O(Δx⁴, Δt⁴) (ν = 1/√3)
Rank-7 (2c+1r, ν = 1, W = 13)
Axis (1, 0, 0)
Face diag (1, 1, 0)
Body diag (1, 1, 1)
(b) Broadband pulse waveforms at face-diagonal receiver (34, 34, 0)
Pressure $p(\mathbf{x}_{fd}, t)$
Propagation distance $c_0 t/\Delta x$ [grid cells]
Exact 3-D D'Alembert (r = 48.08 Δx)
7-pt FDTD (rel L² = 109.3%)
27-pt IWB ν = 1 (rel L² = 98.2%)
Rank-7 2c+1r, ν = 0.75 (rel L² = 2.5%)
Rank-7 2c+1r, ν = 1.00 (rel L² = 2.2%)
(c) 7-pt FDTD (z = 0 slice, $c_0 t \approx 48\Delta x$)
(d) 27-pt IWB (ν = 1, R = 1 real Gaussian)
(e) Rank-7 multi-Gaussian (2c+1r, W = 13)
y/Δx
x/Δx
Pressure p(x, y, 0, t)